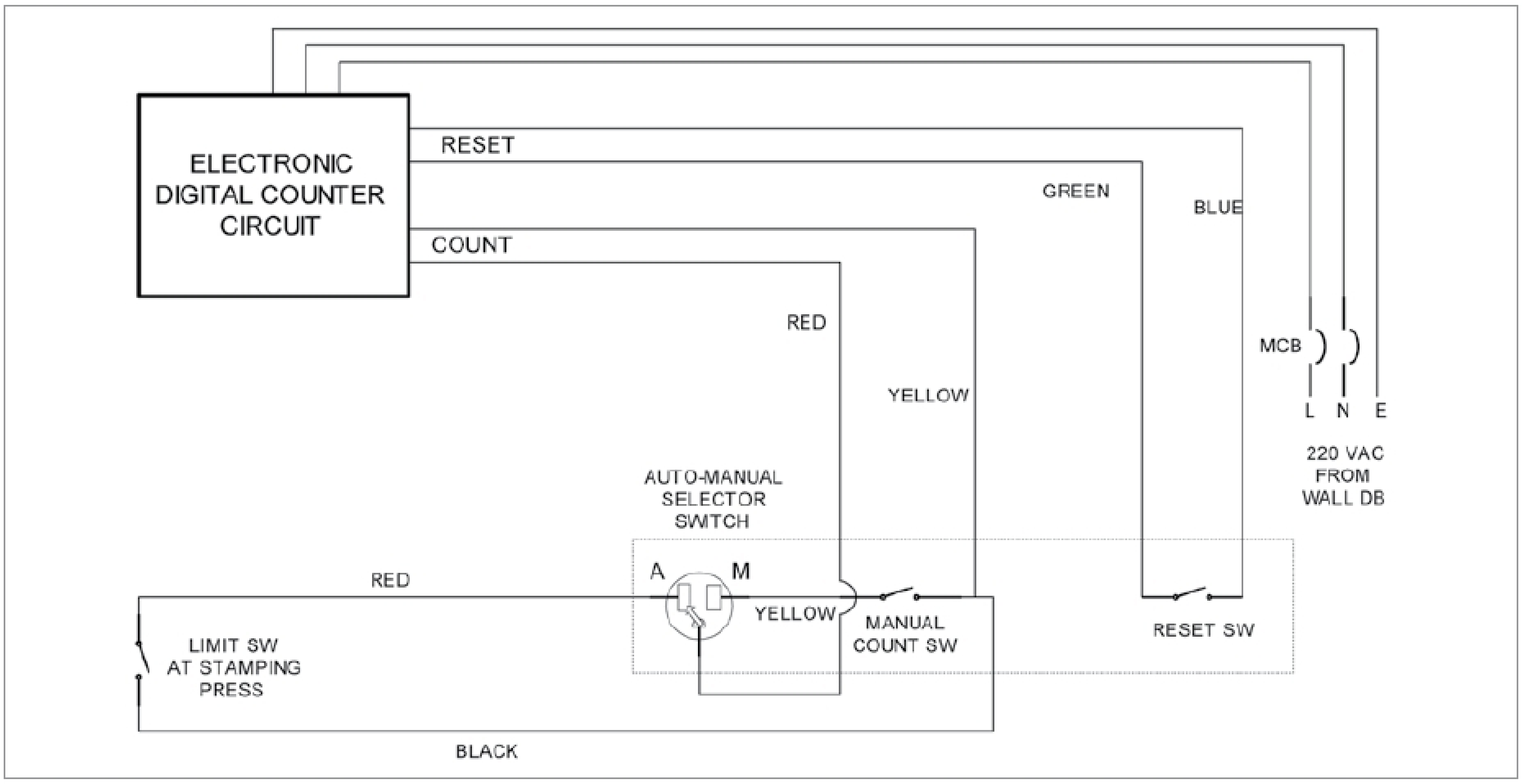

Fig. 7: Electrical wiring diagram of three-digit digital wheel counter

extra production will be more than a single wheel in a shift. Its impact/effect is cumulative. Sensitivity analysis for the extra wheels produced is given in the table.

Schematic layout and the process of wheel forging and wheel milling

The process of wheel forging and rolling is described below.

Forging area and wheel forging process. In the mill providing section of WAP—an upstream unit of wheel forge—required-sized blocks are cut from input ingots received from the steel-melting shop. These blocks are then sent to the entry door area of the furnace (rotary hearth type) through the roller table. The rail charger picks up the blocks one by one and places them in the furnace. From the discharge door, mobile charger G2 takes the soaked blocks and places them on the descaling platform.

After descaling by shaking on the platform, G1 picks up the blocks and places them on the anvil of 63/12MN forging press. In the forging press, the blocks are given the shape of the wheel by punching a hole in the centre in three stages.

After punching, blocks are taken away by G2 at wheel mill side for placing in the furnace on the other side.

Sensitivity Analysis for Extra Wheels Produced

Total extra wheel production in a shift	Total extra wheel production in a day	Envisaged financial benefit (approx.) (in ₹ million)
1	3	60
2	6	120
3	9	180
4	12	230
5	15	290
6	18	350
7	21	410
8	24	470
9	27	530
10	30	590
15	45	880
20	60	1170
25	75	1470
30	90	1760
35	105	2050
40	120	2350

The whole operation is shown in the schematic layout given in Fig. 4.

Wheel mill area and rolling process. Forged wheels are taken from 63/12MN forging-cum-punching press by G2 and placed in the second furnace for re-heating, to raise the rolling temperature for rolling in the wheel mill. From the second furnace, hot wheels are sent to wheel mill by G2 for rolling, and a final shape (profile) is obtained.

Rolled wheels are then taken by mobile charger M1 and put on the dishing press for dishing. Dished wheels are moved to the stamping press by M1 for stamping—this is done for identification of the wheels. After stamping, finished forged wheels are placed on the floor by M1.

M2 moves these hot wheels to the hot bed in stacks of two for cooling. Wheels after preliminary cooling are moved by the crane to the heat treatment area. Entire operation is shown in the schematic layout given in Fig. 5.

Conclusion

The automatic digital wheel counter with jumbo-sized seven-segment displaying system has been working well at the forging shop of W&A Plant for the last two years. Production of forged wheels has increased substantially after installation of the display system. It is helping in harnessing the higher potential of the wheel forge shop.

Even though this digital counter is made for counting the number of wheels manufatured at the wheel plant of Durgapur Steel Plant of SAIL, it can be used anywhere for automatic counting and displaying purposes in a large area. EFY